PLANT-POWERED PERFORMANCE

The Vegan Athlete Nutrition Guide

Complete Amino Acid Coverage Without Meat

EvoFit Meals

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1 Can You Build Muscle on Plants?

The idea that you need animal protein to build serious muscle is one of the most persistent myths in sports nutrition. Science and world-class athletes are proving otherwise every single day.

MYTH #1

"Plant proteins are incomplete and can't support muscle growth."

REALITY

Multiple plant foods (soy, quinoa, hemp, buckwheat) contain all 9 essential amino acids. A 2019 study in the *Journal of the International Society of Sports Nutrition* found no significant difference in muscle gains between plant and animal protein groups when total protein intake was matched at 1.6g/kg/day.

MYTH #2

"You have to combine proteins at every meal."

REALITY

Your body pools amino acids throughout the day. As long as you eat a variety of protein sources across your meals, your body will assemble complete proteins. You do NOT need complementary proteins at the same sitting.

MYTH #3

"No elite athletes are vegan."

🎾

Novak Djokovic

Tennis - 24x Grand Slam Champion

Plant-based since 2011. Attributes his longevity, recovery speed, and sustained dominance to his diet.

🏊

Lewis Hamilton

Formula 1 - 7x World Champion

Vegan since 2017. Reports improved energy, mental clarity, and faster recovery between races.

🏋

Patrik Baboumian

Strongman - World Record Holder

Held the world log lift record at 165kg. Set yoke carry world record (555kg) while fully vegan.

Key Takeaway: Aim for 1.6-2.2g protein per kg bodyweight daily from diverse plant sources. An 80kg athlete needs 128-176g protein/day, which is absolutely achievable on plants.

2 Complete Plant Proteins

These 8 plant foods contain all 9 essential amino acids (histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, valine) making them complete protein sources on their own.

🍥 Soy (Tofu/Tempeh/Edamame)

36g protein per cup (tempeh)

Highest leucine of any plant (key for muscle protein synthesis). Tempeh also provides probiotics for gut health. 18g per 100g tofu.

🌾 Quinoa

8g protein per cup (cooked)

Exceptionally high in lysine, the amino acid most plants lack. Also rich in iron, magnesium, and manganese. A true superfood grain.

🌿 Hemp Seeds

10g protein per 3 tbsp

Ideal 3:1 omega-6 to omega-3 ratio. Rich in arginine (supports blood flow). Easy to add to smoothies, salads, and bowls.

📦 Buckwheat

6g protein per cup (cooked)

Gluten-free despite the name. High in rutin (supports circulation). Excellent as soba noodles, porridge, or flour for pancakes.

🥑 Chia Seeds

5g protein per 2 tbsp

Packed with omega-3 ALA (5g per serving). Absorb 12x their weight in water for sustained hydration. High fiber (10g per serving).

🌱 Amaranth

9g protein per cup (cooked)

Higher in lysine than any other grain. Rich in iron (5mg per cup) and calcium. Excellent popped, as porridge, or in baked goods.

💚 Spirulina

8g protein per 2 tbsp

60-70% protein by weight, the highest of any food. Rich in B12 analogue, iron, and phycocyanin (powerful antioxidant). Add to smoothies.

🧀 Nutritional Yeast

8g protein per 3 tbsp

Fortified with all B vitamins including B12. Cheesy flavor makes it perfect on pasta, popcorn, and sauces. Rich in zinc and selenium.

Pro Tip: For maximum muscle protein synthesis, aim for at least 3g of leucine per meal. Combine soy (highest leucine) with hemp or chia seeds for an amino acid powerhouse. Tempeh + quinoa bowl = roughly 44g complete protein with optimal leucine content.

3 Essential Supplements

A well-planned vegan diet covers most nutritional needs, but these 6 supplements fill critical gaps that are difficult to meet through food alone. Every vegan athlete should consider these non-negotiables.

Vitamin B12

2,500mcg/week (cyanocobalamin)

Best form: Cyanocobalamin sublingual tablet or spray

Critical for nerve function and red blood cell production. No reliable plant source exists. Deficiency causes fatigue, weakness, and neurological damage. Non-negotiable for every vegan.

Iron

18-27mg/day (if levels are low)

Best form: Iron bisglycinate (gentle on stomach)

Plant iron (non-heme) absorbs at 2-20% vs 15-35% for heme iron. Take with vitamin C to boost absorption 3-6x. Avoid taking with calcium, coffee, or tea.

Omega-3 (EPA/DHA)

250-500mg combined EPA+DHA daily

Best form: Algae-derived oil (skip fish, go straight to the source)

ALA from flax/chia converts to EPA/DHA at only 5-10%. Direct algae supplementation ensures adequate anti-inflammatory omega-3s for recovery and joint health.

Vitamin D3

2,000-4,000 IU/day

Best form: Vegan D3 from lichen (not lanolin-based D3)

Essential for calcium absorption, bone density, immune function, and testosterone production. Most people are deficient regardless of diet. Get levels tested annually.

Zinc

15-30mg/day

Best form: Zinc picolinate or citrate

Phytates in grains and legumes reduce zinc absorption by up to 45%. Critical for testosterone production, immune function, and protein synthesis. Take away from iron supplements.

Creatine Monohydrate

5g/day (no loading phase needed)

Best form: Micronized creatine monohydrate

Vegans have 20-30% lower baseline creatine stores than omnivores. Supplementation improves power output, strength, and cognitive function. The most researched sports supplement ever.

Supplement Stack Timing

- ✓ **Morning with food:** B12 + Vitamin D3 + Omega-3 (fat-soluble vitamins need dietary fat)
- ✓ **Post-workout:** Creatine (5g with your protein shake)
- ✓ **Evening with food:** Iron + Vitamin C (separate from zinc by 2 hours)
- ✓ **Before bed:** Zinc (on an empty stomach for best absorption)

Important: Get blood work done every 6-12 months to track B12, iron (ferritin), vitamin D, and zinc levels. Supplement dosages should be personalized based on your lab results, not generic recommendations.

4 7-Day Vegan Meal Plan (2,200 cal)

Daily targets: ~2,200 calories | 140g protein | 260g carbs | 75g fat. Designed for a 75kg athlete in a lean gaining or maintenance phase.

Day	Breakfast	Lunch	Dinner	Snack
Mon	Tofu scramble w/ spinach, mushrooms, nutritional yeast, 2 slices sourdough P: 32g | C: 45g | F: 14g	Chickpea curry w/ brown rice, roasted cauliflower, side salad P: 28g | C: 72g | F: 18g	Lentil bolognese over whole wheat pasta, side of steamed broccoli P: 34g | C: 78g | F: 12g	Protein shake w/ banana, peanut butter, soy milk P: 38g | C: 42g | F: 16g
Tue	Overnight oats: oats, chia, hemp seeds, berries, almond butter, soy milk P: 26g | C: 58g | F: 18g	Tempeh stir-fry w/ bell peppers, snap peas, brown rice noodles, tamari P: 36g | C: 52g | F: 16g	Black bean tacos: corn tortillas, guac, pico de gallo, shredded cabbage P: 30g | C: 68g | F: 20g	Edamame (1 cup) + handful of almonds P: 24g | C: 18g | F: 16g
Wed	Smoothie bowl: pea protein, frozen acai, banana, granola, hemp seeds P: 34g | C: 62g | F: 12g	Quinoa power bowl: roasted sweet potato, chickpeas, tahini dressing, kale P: 26g | C: 72g | F: 18g	Stuffed bell peppers w/ lentils, brown rice, tomato sauce, nutritional yeast P: 30g | C: 64g | F: 14g	Protein bar (vegan) + apple w/ almond butter P: 28g | C: 38g | F: 18g
Thu	Chickpea flour omelette w/ veggies, avocado toast on rye P: 28g | C: 48g | F: 20g	Buddha bowl: baked tofu, edamame, brown rice, pickled ginger, miso dressing P: 38g | C: 58g | F: 16g	Red lentil dal w/ basmati rice, roasted eggplant, naan (vegan) P: 28g | C: 82g | F: 10g	Trail mix (walnuts, pumpkin seeds, dark chocolate, dried mango) P: 14g | C: 32g | F: 22g
Fri	Peanut butter banana protein pancakes (oat flour, pea protein, soy milk) P: 36g | C: 56g | F: 16g	Mediterranean wrap: hummus, falafel, cucumber, tomato, mixed greens P: 24g | C: 62g | F: 20g	Tempeh stir-fry w/ broccoli, cashews, jasmine rice, sriracha P: 38g | C: 64g | F: 18g	Greek-style soy yogurt w/ granola and mixed berries P: 20g | C: 36g | F: 8g
Sat	Loaded avocado toast: sourdough, white beans, cherry tomatoes, hemp seeds P: 22g | C: 42g | F: 22g	Thai peanut noodles: rice noodles, tofu, shredded carrots, cilantro, lime P: 32g | C: 68g | F: 20g	Chickpea curry w/ coconut milk, spinach, basmati rice P: 26g | C: 74g | F: 16g	Protein shake: pea protein, frozen cherries, oats, soy milk P: 40g | C: 48g | F: 8g
Sun	Breakfast burrito: scrambled tofu, black beans, salsa, avocado, tortilla P: 34g | C: 52g | F: 18g	Lentil soup w/ crusty bread, side of quinoa tabbouleh P: 28g | C: 66g | F: 12g	Spaghetti w/ walnut-mushroom meat sauce, garlic bread, side salad P: 30g | C: 72g | F: 20g	Hummus w/ veggie sticks + 2 tbsp pumpkin seeds P: 16g | C: 24g | F: 14g

Daily Average: 2,185 cal | 138g protein | 262g carbs | 74g fat. Protein hits 1.84g/kg for a 75kg athlete, well within the optimal 1.6-2.2g/kg range for muscle growth.

Customization Tip: Need more calories? Add an extra tablespoon of nut butter (+100 cal, +4g protein). Cutting? Remove the snack and reduce rice portions by half (-400 cal).

5 Protein Combining Chart

While combining isn't required at every meal, these classic pairings create complete amino acid profiles by pairing foods that complement each other's limiting amino acids. Grains are low in lysine; legumes are low in methionine. Together, they cover all bases.

Rice High methionine	+	**Beans** High lysine	=	**Complete Protein** ~14g per serving
Pita Bread High methionine	+	**Hummus** High lysine	=	**Complete Protein** ~12g per serving
Whole Wheat Bread High methionine	+	**Peanut Butter** High lysine	=	**Complete Protein** ~15g per serving
Corn Tortillas High methionine	+	**Black Beans** High lysine	=	**Complete Protein** ~13g per serving
Pasta High methionine	+	**Lentil Sauce** High lysine	=	**Complete Protein** ~18g per serving

Other Winning Combos

- ✓ Oatmeal + soy milk + hemp seeds
- ✓ Lentil soup + crusty bread
- ✓ Stir-fry tofu + brown rice
- ✓ Granola + soy yogurt + walnuts
- ✓ Minestrone soup + white beans

Limiting Amino Acids

- ✓ **Grains:** Low in lysine, threonine
- ✓ **Legumes:** Low in methionine, cysteine
- ✓ **Nuts/Seeds:** Low in lysine
- ✓ **Vegetables:** Low in methionine
- ✓ **Soy:** Complete (no limiting AA!)

Remember: You don't need to combine at every meal. Eating a variety of grains, legumes, nuts, and seeds throughout the day naturally provides all essential amino acids. Combining is a bonus, not a requirement.

6 Vegan Meal Prep Tips

Consistency beats perfection. These meal prep strategies ensure you always have high-protein vegan meals ready to go, eliminating the #1 reason athletes fall off their nutrition plan: convenience.

Batch Cook Beans & Lentils

- Cook 2-3 varieties weekly (chickpeas, black beans, lentils)
- Use an Instant Pot: dried beans in 25-40 min, no soaking
- Store in 2-cup portions in fridge (5 days) or freezer (3 months)
- Season differently: Mexican (cumin, chili), Indian (turmeric, garam masala), Mediterranean (oregano, lemon)

Prep Grains in Bulk

- Cook 4-5 cups dry rice/quinoa for the week
- Quinoa: 1:2 ratio, 15 min simmer, fluff and cool
- Brown rice: 1:2.5 ratio, 45 min or rice cooker
- Divide into meal-sized containers immediately
- Freeze extra portions flat in bags for quick reheating

Freezer-Friendly Meals

- **Lentil bolognese:** Makes 8 servings, freezes 3 months
- **Chickpea curry:** Even better after freezing/reheating
- **Bean chili:** Portion into single servings for grab-and-go
- **Veggie burgers:** Shape, freeze on tray, bag once solid
- **Soup portions:** Freeze in muffin tins for perfect sizes

Tofu & Tempeh Mastery

- **Press tofu:** 30 min under weight, then cube and marinate
- **Freeze tofu first:** Changes texture to chewy/meaty
- **Batch bake tofu:** 400F for 25 min, toss in sauce
- **Crumble tempeh:** Sautee with taco seasoning for quick protein
- Marinated tofu keeps 5 days in fridge

High-Protein Smoothie Formulas (30-40g protein each)

Protein Powder (1 scoop pea/soy) + Frozen Fruit (1 cup) + Seeds/Nut Butter (1-2 tbsp) + Soy/Oat Milk (1.5 cups) + Greens (handful spinach)

Chocolate Power: Pea protein + frozen banana + cacao + PB + soy milk (38g protein)
Berry Blast: Soy protein + mixed berries + hemp seeds + oat milk + spinach (34g protein)
Tropical Green: Pea protein + mango + chia seeds + coconut water + kale (32g protein)
Recovery Shake: Pea/rice blend + tart cherry + almond butter + soy milk + creatine (40g protein)

Sunday Prep Checklist (2 hours)

- ✓ Cook 2 types of beans (Instant Pot)
- ✓ Make 4 cups quinoa + 4 cups rice
- ✓ Bake 2 blocks of tofu
- ✓ Prep overnight oats x 3
- ✓ Wash and chop all vegetables
- ✓ Make 1 big batch sauce or curry

Pantry Essentials

- ✓ Canned beans (chickpeas, black, kidney)
- ✓ Dried lentils (red and green)
- ✓ Nutritional yeast (bulk bag)
- ✓ Pea protein powder (unflavored)
- ✓ Tahini, soy sauce, miso paste
- ✓ Hemp seeds, chia seeds, flax meal

READY TO GO PLANT-POWERED?

AI-Powered Vegan Meal Plans Generated in Seconds

Stop guessing macros and Googling recipes. EvoFit Meals builds complete vegan meal plans tailored to your body, goals, and taste preferences using AI.

Starter	Pro Athlete (BEST VALUE)	Team
Free	**$9**/mo	**$29**/mo
5 AI meal plans/month	Unlimited AI plans	Up to 10 athletes
Basic macro tracking	Custom macro targets	Coach dashboard
50+ vegan recipes	500+ vegan recipes	Meal plan sharing
	Supplement tracking	Priority support

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✓ No credit card required ✓ Vegan-optimized AI ✓ Complete amino acid tracking

✓ Auto grocery lists

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